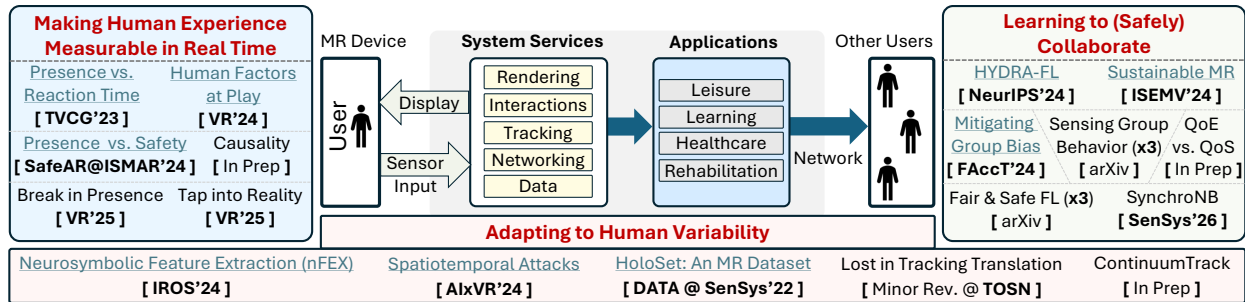


I am a Human-Centered Computing researcher advancing Mixed Reality (MR) to improve how people work, learn, heal, and connect. My research focuses on designing the next generation of adaptive Human-Centered MR by aligning technologies with user experiences and integrating human factors into every aspect of MR design and development, using tools and techniques from Human-Computer Interaction (HCI), Systems, and Machine Learning (ML). I achieve this by measuring and interpreting subjective human experiences and then adapting key system parameters and services in MR environments to enhance user immersion, safety, and overall experience.



Mixed Reality (MR) combines physical and virtual worlds by sensing the user's environment and interactions and using position tracking to align virtual elements with real-world objects accurately. This alignment matters because it allows patients in rehabilitation to see exercises unfold seamlessly, learners to manipulate complex concepts tangibly, and collaborators to share a common workspace that feels natural. The system processes this tracking data so virtual objects respond in real time, giving users experiences that feel both believable and safe. This immersion creates the experiential core of MR, which is presence, the feeling of "being there" in the virtual environment. Presence is both a technical milestone and a human outcome, and its influence depends on the downstream application. Whether therapy motivates recovery, whether training builds confidence, and whether collaboration sustains trust across distance all play a role in its impact. MR applications, therefore, place unique demands on both user experience and the underlying technology, requiring systems that adapt to human variability and contextual needs. For instance, gaming prioritizes responsiveness and visual quality, while surgical applications depend on precise tracking for patient safety. In collaborative settings, systems must also manage what to share and when, balancing resources, privacy, security, and fairness so that no single participant undermines the collective experience.

Given these demands, where MR must handle the unpredictability of human interactions and adapt to changing conditions in environment, behavior, collaboration, and resource use, my work focuses on three thrusts that translate technical design into human impact including (1) creating experiential metrics that capture signals of human state, such as using reaction time to reflect presence and cognitive load [1–6], enabling environments that adapt as users get distracted, fatigued, or overloaded, (2) building adaptive tracking methods that remain reliable across the natural variability of human movement [7, 8], supporting applications such as rehabilitation, where irregular gestures and fatigue are part of recovery, (3) developing collaborative MR sensing systems that passively capture multimodal data to infer group dynamics [9–11], enabling real-time detection of collaboration patterns and supporting safe and fair collaboration [12–15].

① Making Human Experience Measurable in Real Time (VR x 3, TVCG, SafeAR@ISMAR). **Experiential Thrust** asks how MR environments can recognize what people feel and how MR

can adapt to keep them engaged, safe, and included. To do this, I developed experiential metrics that connect human perception, cognitive state, and real-time interaction, since improving user experience requires measuring it accurately. I identified reaction time (the time it takes for a user to respond to a cue) as a key measure of presence (a subjective feeling that reflects the user's immersive experience). Reaction time is readily available and reveals how visual, auditory, and task demands shape the user experience, making it possible to adapt environments in real-time for patients in rehabilitation, students in classrooms, or teams collaborating remotely.

I showed that reaction time is influenced by sensory design and human cognition. By investigating the *link between the human's cognitive state and their performance* [1], I found that factors like familiarity and task conditioning shape how people respond. This demonstrated that reaction time reflects both the fidelity of the scene and the user's cognitive state. By *linking human perception (subjective experiential element) to human actions (systemic performance indicator)* [2], my work gave designers a tool to see how their choices directly affect what users feel and do, allowing training environments to build confidence and therapy tools to adjust difficulty without overwhelming patients. Building on this, I evaluated how interaction design impacts presence, using reaction time as a way to *assess presence (experiential) and fine-tune interactions (systemic) to improve the overall experience and design* [3]. Intuitive interactions that mimic the real world reduced cognitive load and enhanced presence, making systems easier to use for learners and patients. In contrast, abstract interactions increased load, slowed responses, and diminished immersion. These findings emphasized the importance of *designing MR interactions that are both perceptually intuitive and cognitively manageable for users* [5], enabling access across diverse cognitive and physical needs.

Finally, I studied how everyday interruptions disrupt immersion, known as break-in-presence (BIP). Congruent distractions, such as a familiar sound, were easier to rationalize, while incongruent ones caused cognitive overload, longer reaction times, and loss of presence. I quantified this by modeling the *mediation between distraction, cognitive load, reaction time, and presence* [4], showing how disruptive events threaten safety and delay recovery. These insights provide pathways for MR systems that anticipate user vulnerability and adapt to protect users during surgery, therapy, or collaborative work, where a lapse in attention can have lasting consequences.

② Adapting to Human Variability (IROS, AIXVR, DATA@SenSys, LBW@Sensys).

Systemic Thrust contributes to core technological infrastructure, improving pose tracking accuracy, adaptability, and security in MR. In MR, users move fluidly between settings such as indoors and outdoors under varied lighting while interacting with both physical and virtual elements. Tracking failures in these environments is not just about technical glitches; they also add cognitive strain, break immersion, and, in safety-critical environments, compromise physical safety. To address this, I developed Neurosymbolic Feature Extraction (nFEX), an adaptive framework for *human-centric feature extraction for pose tracking* [7]. nFEX combines the clarity of symbolic reasoning with the adaptability of neural networks, enabling systems to respond to changes in lighting, textures, and user motion. To support it, I created two resources: symbolic rules that inform the *design of tracking solutions suited to unpredictable human interactions* [8], and the *HoloSet dataset capturing previously unexplored human interactions in MR* [16]. With ~100K multimodal samples, HoloSet validated nFEX's adaptability, showing up to 90% error reduction in complex scenarios, directly improving safety and usability in rehabilitation, surgery, and collaborative learning.

Even with adaptability, MR systems remain vulnerable to stealthy manipulations. I uncovered a *novel attack surface that exploits the multimodal and spatiotemporal nature of MR tracking systems* [17, 18], demonstrating how subtle manipulations can quietly degrade accuracy and

threaten safety. To complement nFEX's real-time adaptability, I introduced *ContinuumTrack* [19], a system that anticipates and resolves tracking failures before they occur, combining profiling, prediction, and correction to sustain long-term reliability. Together, these advances enable MR that is immersive, trustworthy, and safe for safety-critical domains.

③ Learning to (Safely) Collaborate (*NeurIPS, FAccT, Sensys, eEnergy, ISEMV*).

User Collaboration Thrust focused on fairness, privacy, security, and resource efficiency in shared MR experiences to promote equitable interactions between users. My contributions included developing techniques for multimodal sensing to *sense and interpret individual behaviors in a group to infer group dynamics* [11], such as identifying whether users are collaborating, competing, or working independently. I built a prototype that passively captured user behavior and applied sociometric analysis to infer group activity in real-time [10], determining whether the individual perception of collaboration matches the actual experience [9]. This helps MR environments adapt to collaboration patterns and prevent individuals from being excluded.

However, continuous sensing and readily available data raise security and privacy concerns by potentially capturing sensitive information like private conversations or personal habits. Disproportionate data collection from users with specific speech patterns or gestures may introduce biases, leading to misinterpretation and unfair treatment of some users, skewing collaboration dynamics. Additionally, malicious users manipulating virtual objects could inject biased data, expose private interactions, or cause security breaches, compromising the MR environment. These challenges underscore the need for strong privacy, security, and fairness measures in collaborative MR systems, which my work addresses by protecting sensitive data and supporting fair model training. Through Federated Learning my work *supports secure and fair data handling across multiple users* [12–14, 20–22]. This will enable applications in classrooms and therapy groups where individual support enhances rather than disrupts the shared experience.

Overall, participation and symmetrical collaboration also depend on system performance and sustainability. I studied the *critical trade-offs between quality-of-service (QoS) and quality-of-experience (QoE)* [23] and advanced methods to *minimize the carbon footprint and enhance resource efficiency without sacrificing performance* [24–26]. These advances make collaborative MR both inclusive and environmentally responsible, supporting equitable participation while aligning with broader societal goals.

Future Vision

Current platforms assume neurotypical behavior, treat variability as error, and offer little to no transparency to users. I imagine a future where MR is a vital part of therapy, rehabilitation, and education, with systems that place human experience at the center. Presence, cognitive load, movement variability, and collaboration dynamics serve as signals that guide adaptation, with participation as the outcome. The next phase of my work will move into real-world clinical and educational deployments where adaptive MR can demonstrate impact on therapy, rehabilitation, classrooms, and cognitive health.

① Inclusive Sensing of Cognitive and Emotional States

My current work shows that reaction time can capture presence and cognitive load in real time. I will expand this into richer multimodal sensing pipelines incorporating gaze, voice, and facial expressions to create MR systems that continuously estimate both cognitive and emotional states. This will make immersive environments perceptually, physically, and socially aware.

The **short-term goal** is to design systems that monitor explicit feedback (reaction time) alongside implicit cues (gaze shifts, speech prosody, facial expressions). These prototypes will dynamically adjust immersion and pacing to prevent overload, particularly in health and accessibility-

focused applications where cognitive overload or physical disorientation poses a critical risk. The question I aim to answer is how to account for individual variability in users' responses and tailor safety interventions to meet each user's needs.

The **long-term goal** is to build adaptive behavioral models that learn from both neurotypical and neurodivergent populations. By capturing variations in attention, frustration, or engagement, these systems will tailor therapy, education, and wellness applications to individual cognitive profiles. With deeper interdisciplinary training in cognitive science, neuroscience, and clinical collaboration, I will design therapy environments that adapt strategies dynamically and provide individualized pathways for children with autism, ADHD, and sensory processing differences.

② **Learning from Human Variability** Building on my contributions in adaptive tracking and experiential metrics, I will reframe human variability not as error but as signal to design MR systems that recognize natural diversity in movement, attention, and emotion. Current MR systems often fail when users deviate from expected norms, but these deviations are critical for clinical, educational, and accessibility contexts.

The **short-term goal** is to develop proof-of-concept systems that integrate experiential signals such as reaction time, gaze, and irregular motion into adaptive feedback loops. These prototypes will adjust rehabilitation exercises in response to fatigue, simplify sensory input for neurodivergent users, and sustain balanced participation in classrooms by detecting disengagement early.

The **long-term goal** is to scale this approach with generalizable learning frameworks that evolve with users over time. I will create rehabilitation systems that integrate motor, cognitive, and emotional signals into comprehensive recovery models that evolve with the patient and generalize across diverse conditions. These systems will serve aging populations by modeling variability as an early marker of cognitive decline, support therapy by adapting to individual recovery trajectories, and build trust that MR can flexibly respond to diverse human needs.

③ **Equitable and Symmetrical Collaborative MR**

Collaboration is central to the promise of MR but poses challenges of fairness, bias, and sustainability. My work on multimodal collaboration sensing and federated learning provides the foundation for creating collaborative platforms that preserve trust, fairness, and efficiency. For neurodivergent populations, these platforms can both interpret and explain the interactions of others in clearer ways and adapt to their own experiences in real time, so that group work remains accessible and inclusive.

The **short-term goal** is to design collaborative MR systems that detect participation gaps, mitigate bias, and defend against malicious actors. These prototypes will support fairness in distributed settings, such as group therapy, clinical training, and classrooms, ensuring that no user dominates or is excluded, and that no user can induce malicious interactions in a group setting.

The **long-term goal** is to build scalable, cross-platform MR health and learning platforms that integrate fairness-aware federated models with resource-aware optimization while scaling across devices. In classrooms, I will combine multimodal sensing with fairness-aware federated learning to balance personalization and group coherence at scale, so that students with diverse needs remain included. For aging populations, I will pair experiential signals with causal inference models to identify mechanisms of decline and design adaptive environments that promote independence.

In today's world, a cross-cutting element for my future work will be the integration of conversational generative AI, which can make adaptation transparent and supportive by explaining changes, scaffolding tasks, and offering reassurance. *My long-term vision is MR as a medium that places human experience at its center and creates opportunities for participation across diverse cognitive and physical needs.*

Broader Impact

While my research advances MR systems, it has far-reaching societal implications. The behavioral and safety tools I develop in my experiential and systemic thrusts have applications in accessibility and healthcare, particularly for obesity, diabetes, and aging, and align with the National Institutes of Health (NIH) goals. My real-time behavioral monitoring tools can be applied to monitoring behaviors in neurodivergent individuals or those with autism for personalized support, tracking cognitive and physical activity changes in aging to enable early interventions, and monitoring activity and diet to promote healthier choices in diabetes management¹. Finally, with my interdisciplinary background, I can apply to and find both the NSF and NIH CAREER programs highly relevant to my research trajectory.

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